

Assignment_5

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```
[1]: import numpy as np
     import pandas as pd
     import matplotlib.pyplot as plt
     import seaborn as sns
     from sklearn.model_selection import train_test_split
     from sklearn.preprocessing import StandardScaler
     from sklearn.linear_model import LogisticRegression
     from sklearn.metrics import confusion_matrix, accuracy_score, precision_score, recall_score, f1_score
```

```
[3]: df = pd.read_csv(r'C:\Users\hulaw\Downloads\Social_Network_Ads.csv')
```

```
[5]: df
```

```
[5]:
```

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
..	...	...	...	...	...
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

[400 rows x 5 columns]

```
[7]: df.head()
```

```
[7]:
```

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0

2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0

```
[9]: df = df.drop(['User ID', 'Gender'], axis=1)
```

```
[11]: df.head()
```

```
[11]:
```

	Age	EstimatedSalary	Purchased
0	19	19000	0
1	35	20000	0
2	26	43000	0
3	27	57000	0
4	19	76000	0

```
[15]: X = df[['Age', 'EstimatedSalary']]
      Y = df['Purchased']
```

```
[19]: X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2,
      ↪random_state=42)
```

```
[25]: scaler = StandardScaler()
      X_train = scaler.fit_transform(X_train)
      X_test = scaler.transform(X_test)
```

```
[27]: model = LogisticRegression()
      model.fit(X_train, Y_train)
```

```
[27]: LogisticRegression()
```

```
[29]: Y_pred = model.predict(X_test)
```

```
[31]: cm = confusion_matrix(Y_test, Y_pred)
```

```
[35]: TN, FP, FN, TP = cm.ravel()
      print(f"True Positives (TP): {TP}")
      print(f"False Positives (FP): {FP}")
      print(f"True Negatives (TN): {TN}")
      print(f"False Negatives (FN): {FN}")
```

```
True Positives (TP): 19
False Positives (FP): 2
True Negatives (TN): 50
False Negatives (FN): 9
```

```
[37]: accuracy = accuracy_score(Y_test, Y_pred)
      error_rate = 1 - accuracy
      precision = precision_score(Y_test, Y_pred)
```

```
recall = recall_score(Y_test, Y_pred)
f1 = f1_score(Y_test, Y_pred)
```

```
[39]: print(f"Accuracy: {accuracy:.2f}")
      print(f"Error Rate: {error_rate:.2f}")
      print(f"Precision: {precision:.2f}")
      print(f"Recall: {recall:.2f}")
      print(f"F1 Score: {f1:.2f}")
```

Accuracy: 0.86
Error Rate: 0.14
Precision: 0.90
Recall: 0.68
F1 Score: 0.78

```
[43]: plt.figure(figsize=(6, 4))
      sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=['Not Purchased', 'Purchased'], yticklabels=['Not Purchased', 'Purchased'])
      plt.xlabel("Predicted")
      plt.ylabel("Actual")
      plt.title("Confusion Matrix")
      plt.show()
```

